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RESEARCH ARTICLE

Mapping of the Pharmacological Effect of Prohibited Substances and Methods to Athletic Physiological Characteristics

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Received: 16 April 2025 | **Revised:** 12 September 2025 | **Accepted:** 16 September 2025

Keywords: anti-doping | Olympic Games | physiological requirements | risk assessment | WADA Prohibited List

ABSTRACT

This study presents a novel physio-pharmacological framework for assessing the potential performance-enhancing effects of substances and methods included in the 2022 World Anti-Doping Agency (WADA) Prohibited List across 160 Olympic sport disciplines. An expert panel evaluated 31 consolidated pharmacological categories for their capacity to enhance performance across six core physiological athletic demands. Using a calibrated scoring function and bilinear interpolation, we mapped these effects to each sport's dominant physiological profile. Findings confirmed the well-known ergogenic effects of anabolic agents and erythropoietins for power and endurance-based sports, respectively, including the general ergolytic effects of narcotics and cannabinoids. Moreover, the methodology can be used to evaluate any new sport or substance given their respective physiological demands. The study underscores the need for targeted anti-doping strategies, suggesting that risk assessment and test distribution planning should be aligned with sport-specific physiological demands and the relative performance advantages conferred by different doping strategies. This summary of the current physiological knowledge and pharmacological knowledge has the potential to enhance detection efficiency, optimize resource allocation, and refine prohibited substance screening in elite sport.

1 | Introduction

Organized sport began around three thousand years ago in ancient Greece. There is evidence that the athletes from this period used dietary modifications and natural substances in order to enhance athletic performance. The modern era of sport began in the late 19th and early 20th centuries with the formation of sports clubs and societies and the development of stadia in which the public could watch events. During this period, there were reports of the use of substances to enhance performance. However, more widespread attempts at performance enhancement began in the 1950s and 1960s with the availability of more potent and selective drugs from an emerging international pharmaceutical industry.

In 1967, the International Olympic Committee (IOC) formed its Medical Commission, with its guiding principles being the protection of the health of athletes, respect for medical and sport ethics and equality for competing athletes [1]. In the same year, the IOC published the first list of prohibited substances, which at the time only included stimulants and narcotic analgesics, reflecting the variety of the drugs being used at that time and the limitations of the testing and analysis procedures available. This list of prohibited substances and methods has been reviewed and extensively expanded to include a wide range of substances known to improve performance or be detrimental to the health of the athlete [2]. The Prohibited List is now defined and maintained by the World Anti-Doping Agency (WADA) [3].

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WADA updates the Prohibited List annually, based on criteria defined by the World Anti-Doping Code [4]. The criteria are that the substance or method (1) has the potential to enhance or enhances sport performance, (2) represents an actual or potential health risk to the athlete, and (3) violates the spirit of sport.

Each of the classes of substances and methods that are listed in the Prohibited List, with the exception of beta-blockers, are prohibited in all sports. However, given the wide differences in pharmacology, the effect of a particular substance or method to produce performance enhancement will vary between sports. This will depend on the mode of action of the substance or method and on the physiological characteristics of the athlete that are deemed necessary in order to achieve optimal performance within that sport.

In 2022, the results of a study were published in which the principal aim was to provide a quantitative assessment of the athletic demands across all summer and winter Olympic sport events [5]. The results presented the dominant physiological athletic characteristics needed to excel in these events.

The aim of this study was for an independent panel of experts to assess the potential impact, both positive and negative, of each of the classes of substances and methods on the 2022 WADA Prohibited List on the physiological parameters that influence performance in sport. This aimed to determine which specific prohibited substances and/or prohibited methods were most likely to enhance performance or have a detrimental impact on performance across a range of sports with different physiological demands. This would potentially inform a more strategic and targeted approach to anti-doping testing focusing on detection of the most likely substances and methods to be used to improve performance in a particular sport.

2 | Methods

2.1 | Data Collection

A panel of eight experts was convened to assess the potential performance-enhancing effects of each pharmacological category listed in the 2022 WADA Prohibited List [3]. Experts were recruited based on their professional experience in drug use in sport and included practicing sports medicine physicians, sports-specialist pharmacists, and academic researchers. Their professional affiliations spanned primary and secondary healthcare sectors as well as national and international anti-doping organizations. Panel members had significant expertise in the pharmacological and clinical aspects of drug use and misuse in sport, with many actively advising elite athletes. The panel was geographically diverse, including representatives from Australia, North America, Europe, and the Middle East, and was gender-balanced.

The substances and methods in the WADA Prohibited List were consolidated into 31 pharmacological categories based on their primary mechanism of action. Substances with similar or analogous expected performance effects were grouped together. Categories with recognized gender-specific effects were evaluated separately for each gender. Gender differences were only considered where it was expected that biological responses would be dependent upon the gender of the athlete. Experts were instructed to rate each category based on its theoretical performance-enhancing potential when administered at an optimal dose and timing. Consideration was given to both short-term and long-term administration strategies (e.g., during competition or off-season use). A 5-point Likert scale was used to evaluate the degree of advantage each category could potentially provide across six previously defined [5] physiological athletic demands (Table 1).

Assessment scores were generated from the experts' academic knowledge and practical experience within the field of pharmacology. Panel members were requested to conduct literature searches or to discuss with a more experienced practitioner of clinical prescribing to support their findings, where appropriate. Panel members were provided with a precise definition of the criteria for the six physiological athletic demands. This supported understanding and reduced ambiguity between expert members. Ratings were submitted independently and anonymously by each panel member. A secondary anonymous review phase was conducted to enhance inter-rater agreement; no significant changes resulted, indicating consistency and confidence in the experts' assessments.

Where there was discrepancy, a conference call was arranged between the experts who provided a conflicting result to agree on a consensus assessment score. One category, M3 (gene and cell doping), showed a high degree of disagreement among experts, likely due to limited current evidence [6], and the broad range of technologies and therapies it encompasses. As such, this category was excluded from subsequent analyses.

2.2 | Panel Aggregation and Reliability

All expert ratings per combination of prohibited substance/method and athletic variable were combined by simple majority, eventually taking the mean to break the ties.

Other aggregation methods were explored; none introduced significant changes in the post-aggregation ratings.

Consequentially, each prohibited substance/method was thus characterized by six (aggregated) athletic ratings. Inter-rater reliability was assessed using Krippendorff's α coefficient, a standard measure of agreement ranging from 0 (no agreement between experts) to 1 (perfect agreement) [7].

TABLE 1 | Scale of performance effect.

Score	Level of conferment of advantage
0	Negative effect on performance
1	No advantage
2	Weak potential advantage
3	Moderate potential advantage
4	Strong potential advantage

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2.3 | Physio-Pharmacological Scoring

2.3.1 | Scoring Definition

The primary objective was to quantify the expected performance effect of each prohibited substance or method of doping within each of the 160 Olympic sport disciplines, consistent with a previously established framework [5]. Each sport was mapped to one of six core physiological performance domains.

The overall expected benefit B_{ij} for each sport was computed using the following formula:

$$B_{ij} = \sum_{k=1}^{6} R(s_{ik}, p_{jk}),$$

where i, j, and k are the indexes of the discipline, prohibited substance/method, and physiological variable respectively, s_{ik} is the aggregated athletic demand score per sport i, p_{jk} is the aggregated performance rating for substance/method j, and R is a function that assigns the score for the specific pair of ratings. A positive B_{ij} indicates that the prohibited substance/method j has an expected positive effect for an athlete competing in discipline i. An example of the calculation of a B_{ij} is shown in Table 3.

The function R is defined from first principles outlined in Section 2.3.2: These include a calibration step, in order to maximize the discrimination between pairs of disciplines and prohibited substances/methods.

An alternative formula was also considered, using only the maximum expected benefit across all athletic demands: $B^*_{ij} = \max_{k=1}^{6} R(s_{ik}, p_{jk})$.

2.3.2 | Scoring Function Elicitation

Step 3 uses the scoring function R that assigns a numeric score for each possible combination of athletic rating s_{ik} and pharmacological rating p_{jk}. As a reminder, the s_{ik} range from 1 (no athletic demand) to 8 (extreme athletic demand), while the p_{jk} span from 0 (negative effect) to 4 (strong beneficial effect), with 1 indicating a null effect.

Since the scoring function R is completely arbitrary in both shape and range, we have elicited it from the following set of principles:

1. Its values range from $P = -8$ (extreme performance cost) to $M = 8$ (extreme performance advantage),
2. The value 0 represents a neutral value (no effect of the prohibited substance/method on the athletic demand).

The function is then partially specified by considering the hypothetical extreme scenarios:

1. The highest athletic requirement with the highest potential advantage was assigned the score of $M = 8$,
2. The highest athletic requirement with a null potential advantage was assigned the score of 0,
3. The lowest athletic requirement with highest potential advantage was assigned the score of 0,
4. The lowest athletic requirement with a null potential advantage was assigned the score of $m = 1$.

The combinations of prohibited substances with negative effects on performance were graded separately:

1. The lowest athletic requirement with a negative potential advantage was assigned the score of $p = -4$,
2. The highest athletic requirement with a negative potential advantage was assigned the score of $P = -8$.

Since all the extremes are now well defined, the rest of the scoring surface can finally be determined using bilinear interpolation [8]. This is the simplest method that guarantees a smooth transition between all possible athletic/pharmacological combinations of ratings. Simultaneously, this allows us to assign a score to non-integer ratings that may have been introduced by the aggregation steps.

The specific values for m, M, p, and P were chosen to maximize the discrimination between pairs of disciplines and prohibited substances/methods. For the interested reader, the details are given in the next Section 2.3.2.1.

2.3.2.1 | Calibration of the Scoring Function. The specific values for the parameters m, M, p, and P were chosen by sweeping the four-dimensional parameter space, each ranging from −8 to 8, then repeating the analyses given in Sections 2.3.1 and 2.3.2, and collecting the sets of benefits B_{ij}. The search space was initially reduced by rejecting the combinations that fail these assumptions:

- $M \geq m > 0$: m and M must be positive,[1] as the prohibited substance/method has a potential performance benefit,
- $P \leq p < 0$: p and P must be negative, as the prohibited substance/method has a potential performance cost,
- $|P| = |M|$: A strong potential performance advantage cancels out a strong potential performance cost.

For each chosen combination of m, M, p, and P, a candidate set of benefits $\hat{B}_{ij}$ can be computed using Section 2.3.1. Since the goal of the article is to discriminate between combinations of disciplines and prohibited substances/methods, for each m, M, p, and P, we compute the overall mean and standard deviation of the $\hat{B}_{ij}$:

$$\mu = \frac{1}{N_s N_p} \sum_{i=1}^{N_s} \sum_{j=1}^{N_p} \hat{B}_{ij},$$

$$\sigma = \sqrt{\frac{1}{N_s N_p} \sum_{i=1}^{N_s} \sum_{j=1}^{N_p} \left(\hat{B}_{ij} - \mu\right)^2},$$

where N_s and N_p are the number of sports and prohibited substances/methods, respectively. Finally, to normalize for common

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scaling factors between the parameters m, M, p, and P, we choose the ones that maximize the coefficient of variation $\mathrm{CV} = \frac{\sigma}{\mu}$.

2.4 | Software

All statistical analyses were conducted using R 4.4.1 [9] with the base environment and the tidyverse packages included in version 2.0.0 [10]. Krippendorff's α coefficient was calculated using the R package irr version 0.84.1 [11]. The bilinear interpolation was performed using the R package pracma version 2.4.4 [12].

3 | Results

3.1 | Data Collection and Panel Aggregation

Table S2 in Appendix S1 contains an overview of the 31 considered prohibited substances/methods. Each row was rated independently by 8 experts over the 6 athletic demands. One expert was not able to assess the scores for one prohibited method: This contribution was omitted. The final collected panel data consisted in $6 \times 31 \times 8 - 6 = 1482$ ratings.

All experts perfectly agreed on 15% of the items, and 54% had a maximum difference lower or equal than 1 Likert scale points. The collected dataset was then aggregated according to simple majority voting described in Section 2.2. The outcomes are summarized by substance/method class in Table 2: Just the prohibited substances/methods with a positive effect on performance are shown. The full list after aggregation is available in Appendix S1 in Table S1.

3.2 | Panel Aggregation and Reliability

We measured the rater agreement over 1482 items using Krippendorff's α. Bootstrapping on 2000 iterations, Krippendorff's α lies in the 95% CI [0.49, 0.61], with a mean value of $\alpha = 0.55$ [13]. The obtained value of α indicates a low agreement between experts. Nevertheless, it is slightly larger than the rater agreement of our first study, where the mean value of $\alpha = 0.49$ was obtained [5]. Furthermore, a sensitivity analysis showed no significant changes with respect to the aggregation method (e.g., the mean or the median).

3.3 | Physio-Pharmacological Scoring Function

After applying the calibration procedure described in Section 2.3.2.1 to the aggregated data, the following values for the performance parameters were retained: $p = -4$, $P = -8$, $m = 1$, and $M = 8$. The resulting scoring function R, after interpolation over a fine grid, is shown in Figure 1.

These values were afterwards exploited to compute the effective performance benefits B_{ij} following the equation in Section 2.3.1. An example of calculation for the usage of *EPO* in *Archery* is shown in Table 3.

3.4 | Evaluation of the Physio-Pharmacological Performance Benefit

The set of the B_{ij} is the main outcome of this article and contains all information to assess the advantage of using a prohibited substance/method to compete in a certain discipline. Interested parties include the athletes themselves, whose discipline is set in advance, or anti-doping organizations, who might want to rank sports in terms of overall doping risk, or anti-doping regulators, who might refine the official WADA Prohibited List on a substance/method basis.

3.4.1 | Ranking by Substance

As an example, we might be interested to rank the performance benefit for three selected disciplines (*Archery*, *Athletics: Road - Marathon*, *Weightlifting*) across all evaluated prohibited substances/methods. These sports were chosen as examples solely due to their very different physiological requirements, that should result in different physio-pharmacological performance benefit profiles. Figure 2 shows the relative expected benefits, superimposed over the all-discipline distributions. Our methodology shows that the vast majority of prohibited substances/methods are expected to induce a non-negative effect on the sporting performance. The *Narcotics* and *Cannabinoids* constitute an exception, and *beta-blockers* appear to be helpful only for a subset of disciplines (*Archery* included).

The spread is also variable across drugs: Some of the prohibited substances/methods appear to have a limited effect for all evaluated disciplines, such as *Diuretics*. This is expected, as they are forbidden not for their performance-enhancing effects, but for the masking potential of other prohibited substances.

3.4.2 | Ranking by Discipline

The view of Figure 2 can be reversed by slicing over each discipline and highlighting some of the prohibited substances/methods (*Beta-blockers*, *EPO*, *GH*, *Narcotics*). The substances/methods were selected on the same grounds of the example disciplines in the previous section. To further simplify the visualization due to the large number of disciplines, only a small subset of those is shown. The results are shown in Figure 3.

4 | Discussion

The findings from this study outline the primary perceived physiological advantages or disadvantages conferred from the use of the spectrum of physiological categories of the WADA Prohibited List [3] across 160 Olympic sport events, for the intention of doping. These advantages relate to how the substance or method of doping strategy may improve performance across six athletic characteristics. It was assumed that these doping strategies would be employed at an optimal timing to confer maximum performance enhancing effects, which may be during the preparatory or recovery period outside of competition, or during the competition period itself.

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TABLE 2 | The most beneficial prohibited substances/methods per athletic demand, sorted in decreasing order of the aggregated performance benefit p_{jk}, rounded up.

Aerobic endurance	Muscular endurance	Psychomotor skill and accuracy	Strength and power	Reaction time	Movement agility
Strong potential advantage: $p_{jk} \in (3,\ 4]$					
M1:	**M1**:	**P1**:	**S1**:		
BldRein	BldRein	BB	AAS		
BldO2Enh	**S1**:		SARMs		
S2:	SARMs		**S2**:		
EPO			CGRFs (M)		
			LHRFs (M)		
Moderate potential advantage: $p_{jk} \in (2,\ 3]$					
S4:	**M1**:	**S6**:	**S1**:	**S6**:	
TMZ	BldO2Enh	STM	CZZ	STM	
	S1:		**S2**:		
	AAS		GHRFs		
	S2:		GFs		
	EPO		**S4**:		
	GHRFs		AI (M)		
	GFs		SERMs (M)		
	S4:		ActRIIB-I		
	AI (M)		INS		
Weak potential advantage: $p_{jk} \in (1,\ 2]$					
M2:	**M2**:		**M1**:		**S2**:
BldChem	BldChem		BldRein		CRFs
S1:	**S1**:		BldO2Enh		**S6**:
SARMs	CZZ		**M2**:		STM
S2:	**S2**:		BldChem		
LHRFs (M)	CGRFs (M)		**S2**:		
CRFs	LHRFs (M)		EPO		
CGRFs (M)	CRFs		**S3**:		
S4:	**S3**:		β2-ago		
AI (M)	β2-ago		**S4**:		
SERMs (M)	**S4**:		AMPK-A		
ActRIIB-I	SERMs (M)		SERMs (F)		
AMPK-A	ActRIIB-I		**S6**:		
THP	AMPK-A		STM		
INS	INS		**S9**:		
S6:	THP		GC		
STM	**S6**:				
S9:	STM				
GC	**S9**:				
	GC				

Note: Prohibited substances/methods with no positive effects are omitted.
Abbreviations: β2-ago: beta-2 agonists, (M): males only, (F): females only, AI: aromatase inhibitors, AMPK-A: AMPK activators, ActRIIB-I: ActRIIB inhibitors, BB: beta-blockers, BldChem: blood manip. (chem/phy), BldO2Enh: blood manip. (delivery), BldRein: blood manip. (transf), CGRFs: CG and releasing factors, CRFs: corticotrophins and releasing factors, CZZ: clenbuterol, zeranol, zilpaterol, GC: glucocorticoids, GHRFs: GH and releasing factors, INS: insulin, LHRFs: LH and releasing factors, STM: stimulants, THP: meldonium, TMZ: trimetazidine.

The advantages or disadvantages reported were the independent opinions of experts within the field of doping pharmacology and matched with the athletic demands of 160 Olympic sport events as assessed by [5]. The findings from this study are to assist the anti-doping community in better understanding the primary doping substances that are most at risk of exploitation based upon the purpose of improving performance alone.

4.1 | Sport Event-Specific Doping Strategies

In consideration of the results reported in our former article [5], we considered *Weightlifting*, *Marathon running*, and *Archery* as practical examples of sport disciplines, which rely heavily on one overriding athletic characteristic and the combined primary performance enhancing pharmacological interventions.

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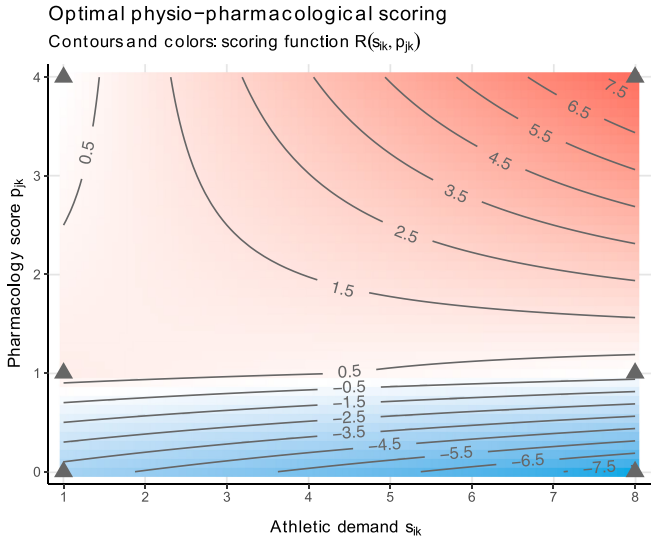

FIGURE 1 | The scoring function $R(s_{ik}, p_{jk})$ after bilinear interpolation over a fine grid. The triangles are placed in the hypothetical extreme scenarios described in Section 2.3.2: In these points, the scoring function is anchored to the known values M, P, m, p, and 0.

TABLE 3 | Calculation of the individual and total performance benefits for one selected sport/drug pair: EPO in Archery.

Athletic variable k	Athletic demand s_{ik}	Potential benefit p_{jk}	Score $R(s_{ik}, p_{jk})$
Strength and power	3.0	1.5	1.00
Muscular endurance	2.2	3.0	1.18
Aerobic endurance	1.2	4.0	0.21
Movement agility	1.4	1.0	0.95
Reaction time	1.4	1.0	0.95
Psychomotor skill and accuracy	8.0	1.0	0.00
Total benefit (sum)			4.29

Weightlifting was assessed as a sport relying maximally on the demands of strength and power with a relatively low physiological demand across other physiological characteristics. As shown in Figure 2, anabolic androgenic steroids, SARMs, and chorionic gonadotrophin [males] were all assessed to have maximal overall expected performance-enhancing benefits. Thereafter, drugs within S1, S2, and S4 [males] all had potential performance-enhancing benefits at a slightly reduced magnitude. From an anti-doping testing perspective, these substances above would be of primary consideration and most at risk of being abused from a physiological standpoint. Unlike endurance sports, Figure 2 also illustrates negligible benefits from blood manipulation or erythropoietin application. For weightlifting, the expected performance benefits of prohibited substances and methods were reviewed in a study by Kolliari-Turner [14] where the most frequently detected substances in sanctioned weightlifters were predominantly anabolic androgenic steroids, markers of endogenous anabolic steroid usage, and other anabolic agents, although a small number of cases of the stimulant methylhexanamine were recorded. On the other hand, reports of EPO use were not described. The assessment by the panel of experts described a broad spectrum of potential benefits on performance in the sport of weightlifting; although, again, most classes of substances and methods were deemed to have no performance-enhancing benefits in this sport.

There is substantial research evidence that erythropoietin has a beneficial effect on athletic performance in many endurance sports [15] and, in particular, in long-distance running events [16]. This was reflected in the assessment results from the panel of experts. Marathon running and other endurance sports low in technicality but high in aerobic endurance characteristics have a maximal overall expected performance benefit from blood manipulation and erythropoietin doping. The anti-doping testing attention should be oriented in the direction of detecting these primary doping strategies.

Archery was the final example within Figure 2. Psychomotor skill and accuracy were, of course, of prime importance in many aiming sports where the other athletic characteristics were considered considerably less important compared to other sports. There are few research publications on the use of performance-enhancing drugs in archery, where hand and body steadiness are key factors. In the 1980s, improvements in shooting performance by beta blockers were attributed to cardiovascular and other effects of beta blockers [17]. This led to the inclusion, in 1985, of beta blockers on the IOC Prohibited List. The current WADA Prohibited List prohibits the use of beta blockers in particular sports, where the control of posture and anxiety play an important role. However, subsequent studies on the benefit of beta blockers on performance have been inconsistent [18, 19]. Indeed, Derman [20] recommended caution in the use of beta blockers by athletes as their effects on exercise are detrimental to competition. This profile of pharmacological effects was reflected in the results of assessment by the panel of experts, where all prohibited substances and methods were deemed to have no performance-enhancing benefits in archery and, in some cases, a negative effect on performance.

4.2 | Substances and Methods With Ergolytic Effects

Not all the WADA prohibited substances or methods listed were recognized to produce performance enhancing benefits. In female athletes, S2.2.1 (chorionic gonadotropin and luteinizing hormone) and S4.1 (aromatase inhibitors) were found to offer no advantage. More notably, several substances, such as S7 (narcotics) and S8 (cannabinoids) were associated with negative impacts on physiological demands such as reaction time, accuracy, and psychomotor skill. Beta-blockers were also detrimental to muscular or aerobic endurance. According to [4], a substance may be also added to the list if it "represents an

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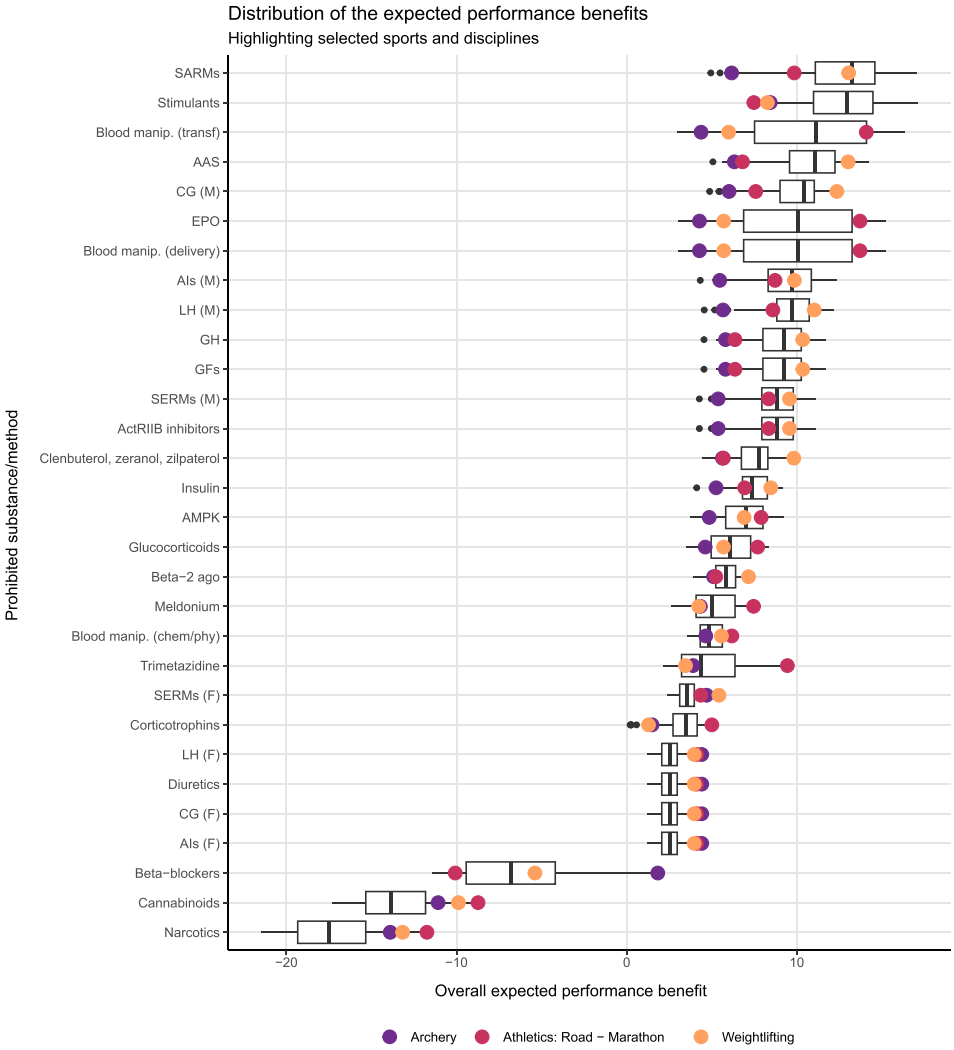

FIGURE 2 | Expected performance benefits for each prohibited substance/method. Several disciplines are highlighted as an example. The *y*-axis is ordered by the median benefit score.

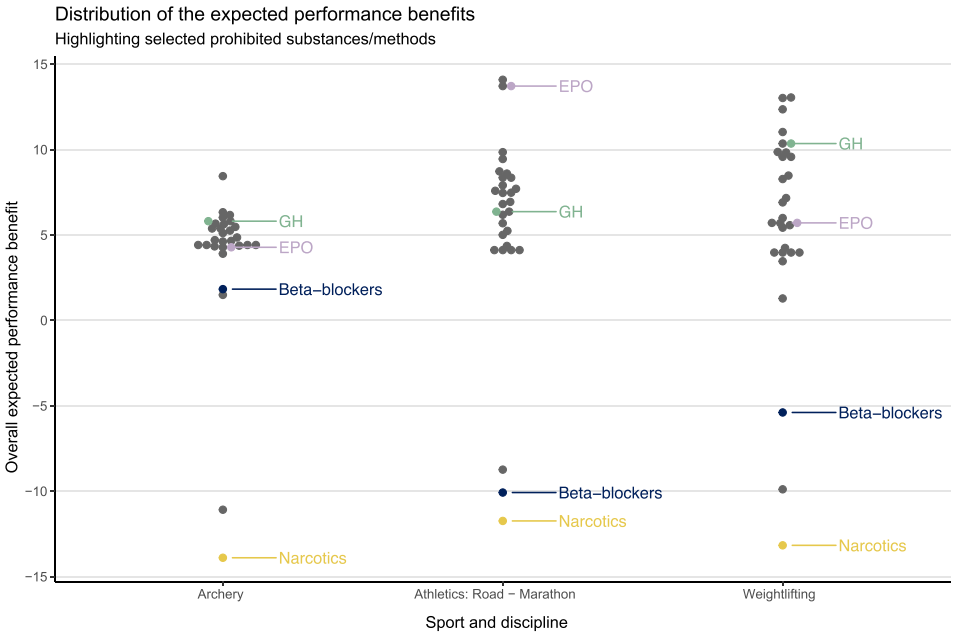

FIGURE 3 | Expected performance benefits for each sporting disciplines. Several are highlighted as an example.

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actual or potential health risk to the athletes" or "violates the spirit of the sport." We therefore acknowledge that some substances are included because of these criteria despite exhibiting no expected benefit to performance, or even an ergolytic effect. This suggests that the inclusion of opioid narcotics and cannabinoids is primarily driven by their well-documented health risks, aligning with WADA's broader criteria beyond just performance enhancement.

4.3 | Risk Assessment and Test Distribution Plan

A well-considered doping risk assessment by an International Federation or major event organizer should include a detailed understanding of the physiological demands of each sport. These demands vary significantly between disciplines. The substances or methods that pose the greatest risk are those offering the highest ergogenic benefits per event.

Test distribution planning (TDP) should therefore prioritize resources toward detecting the substances or methods most likely to be misused, based on their relevance to the event's physiological demands. The results of this study can support risk assessment efforts by highlighting, which substances or methods offer the most performance enhancement per discipline.

The TDP should be primarily shaped around these findings, with the allocation of anti-doping controls and scope of targeted analysis weighted accordingly. The foremost objective is to test athletes at the right time of doping with the right sample type (urine, whole blood, serum or dried blood spot [DBS]), whether in pre-event preparation or competition period, while considering the most effective analytical detection window of the various substances and methods.

4.4 | Limitations and Future Research

This study primarily investigated the physiological performance benefits of prohibited substances and methods in relation to the principal athletic demands of each Olympic sport. Comparing these findings with International Federation testing data, including the quantity and the types of samples collected, was beyond the scope of this research. It is important to acknowledge that some doping practices, despite this study suggesting they offer no ergogenic advantage, have nonetheless led to adverse analytical findings.

There are several indirect factors which may explain why a non-performance enhancing prohibited substance or method might be misused within a sporting context. Diuretics, for example, may be used primarily to act as a masking agent of another prohibited substance. A masking agent can reduce biological abnormalities within the athlete biological passport and limit the analytical capacity of the laboratory to detect the origin of the exogenous prohibited substance. Athletes in weight-class sports may also use them for rapid weight and body-mass reduction to qualify for a weight classification and to gain a competitive edge. Additional motivations may include cultural beliefs, placebo effects, or a fundamental lack of knowledge of the effects of a doping agent has perceived competitive advantage. Some doping strategies may exploit assumptions that certain substances may not be routinely screened for due to limited relevance in terms of performance benefits for specific sporting events. Supporting performance improvement through secondary benefits, such as improved recovery from training or competition, has also been a reason for exploitation. These represent some indirect motivations for doping outside of direct physiological performance enhancement, but by no means constitutes an exhaustive list of all doping tactics and remain outside of the scope of this study.

Future research could investigate the correlation between adverse analytical findings per sport and the perceived doping strategy advantages presented in this study. This task is by no means trivial, as the published statistics are influenced by a large number of mostly unseen confounders, such as the composition of the laboratory testing menu, the type of the collected sample, annual changes to the WADA Prohibited List, the variation of the analytical thresholds for reporting, the quality of the anti-doping tests, and the selection criteria of tested athletes. In addition, one should also account for the sport- and country-specific doping prevalence, as well as the local or online availability of the prohibited substances.

5 | Conclusions

The findings of this study aim to assist the anti-doping community in better identifying the prohibited substances or methods that provide the greatest ergogenic physiological enhancement across Olympic sports. One would expect that these are likely the substances or methods most at risk of being exploited by the athletic community to gain a competitive advantage.

These insights are informed in combination with the evaluated athletic demands across the spectrum of the 2020 summer and 2022 winter Olympic Games sports events [5]. Expert panel scores were mapped to the primary sport-specific athletic demands to provide a physio-pharmacological based scoring system.

This research identifies the primary doping strategies which are most at risk of being abused by sport discipline, based on performance enhancing potential. These findings should inform the development of more targeted and well-considered risk assessments of doping across specific sport disciplines. The anti-doping community is encouraged to consider these findings when refining their test distribution plans. The test distribution plan should be orientated to primarily attempt to detect these substances or methods of doping through the various anti-doping strategies currently available. Orientating anti-doping efforts towards identification of the prohibited substances or methods reported within this research will help strengthen deterrence and improve detection. The findings from this research convey an evidence-based rationale to test for a reduced menu of substances/methods of doping relevant to the perceived advantage per sport discipline. Following this approach will save anti-doping resources and avoid an unnecessary adverse analytical finding from a substance or method that provide no competitive advantage.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The full dataset is available on request.

Endnotes

[1] The choice $m = 0$ is excluded since, for several prohibited methods/substances (e.g., *CG* and *LH* for women), B_{ij} would be 0 for all disciplines irrespective of their athletic demands. Although not wrong, this is undesirable with respect to the article's goal.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Appendix S1:** Supporting information. **Table S1:** The list of considered prohibited substances/methods, and the aggregated potential performance advantage scores across the panel members. **Table S2:** The full list of expected performance benefits, for every combination of prohibited substance/method and sporting discipline.

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